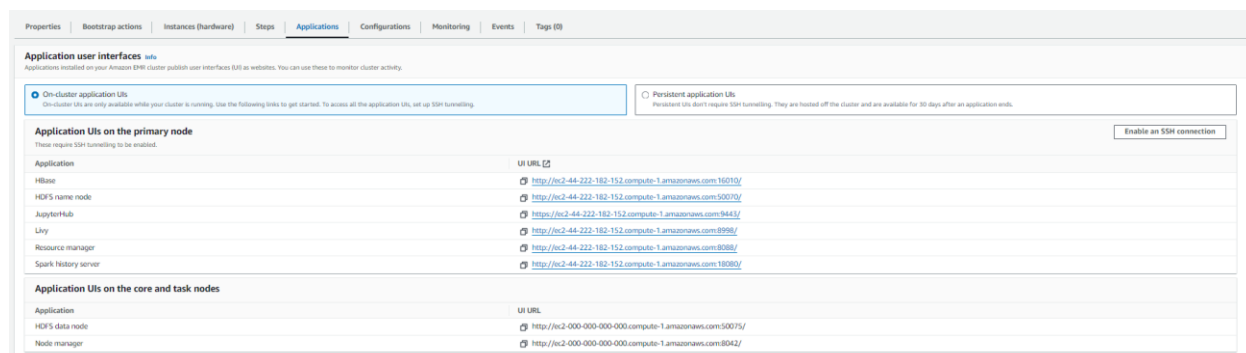
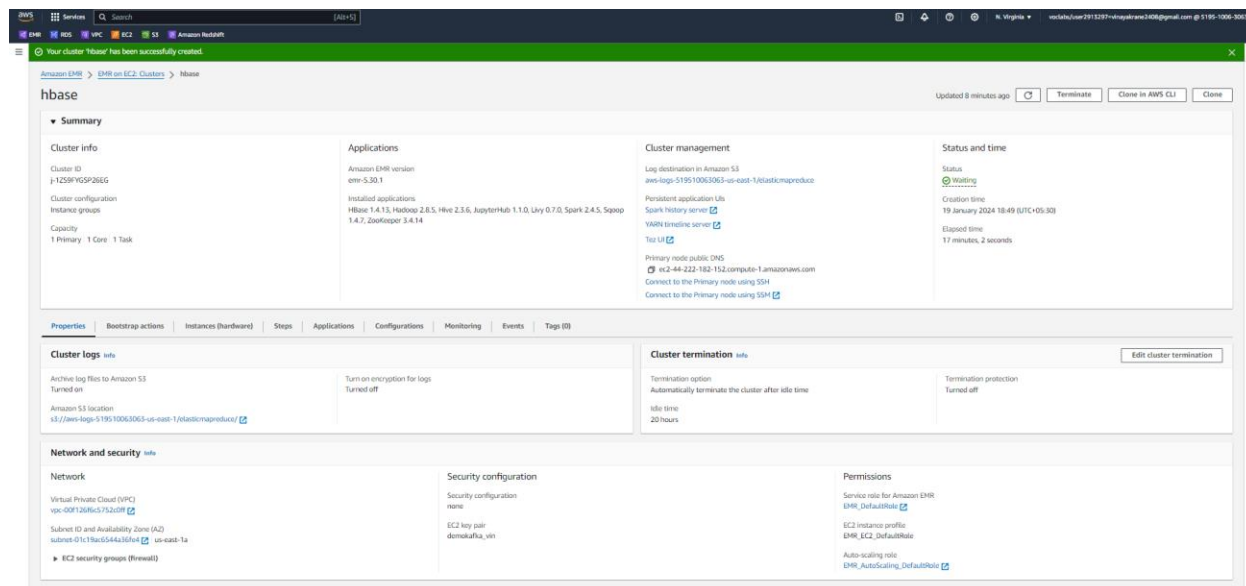


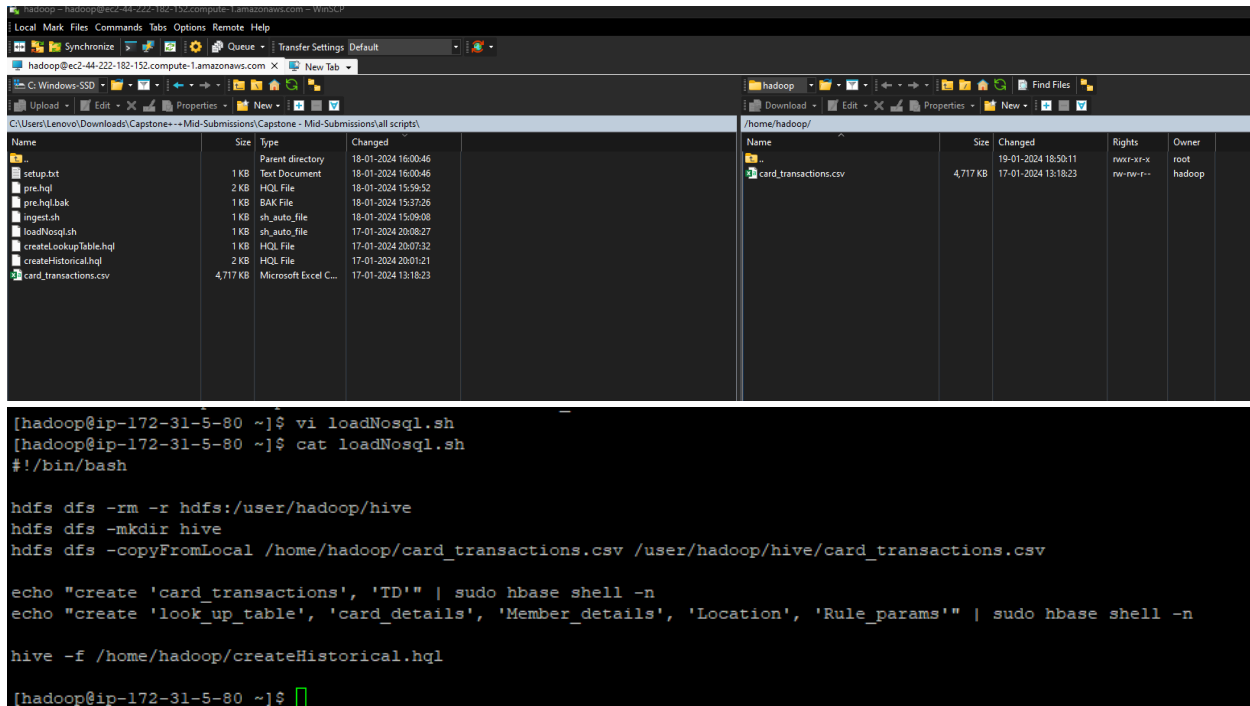
## Scripts Execution

- Screenshot of the EMR Cluster created.



## Steps & Screenshots of the execution of the scripts:

- **Task 1:**  
Copy the **card\_transactions.csv** file from local machine to EMR cluster hadoop instance.



The screenshot shows a terminal window with the following content:

```

[hadoop@ip-172-31-5-80 ~]$ vi loadNosql.sh
[hadoop@ip-172-31-5-80 ~]$ cat loadNosql.sh
#!/bin/bash

hdfs dfs -rm -r hdfs://user/hadoop/hive
hdfs dfs -mkdir hive
hdfs dfs -copyFromLocal /home/hadoop/card_transactions.csv /user/hadoop/hive/card_transactions.csv

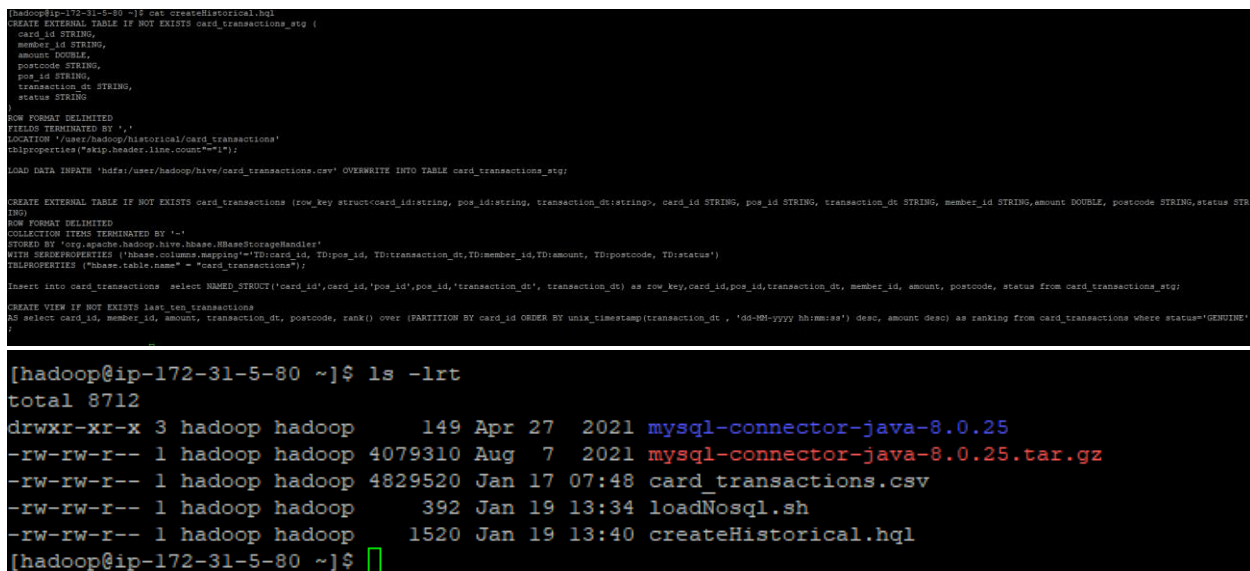
echo "create 'card_transactions', 'TD'" | sudo hbase shell -n
echo "create 'look_up_table', 'card_details', 'Member_details', 'Location', 'Rule_params'" | sudo hbase shell -n

hive -f /home/hadoop/createHistorical.hql

[hadoop@ip-172-31-5-80 ~]$

```

- Load the transactions history data (card\_transactions.csv) in a NoSQL database.
- Script **loadNosql.sh** and **createHistorical.hql** created to complete 1<sup>st</sup> task.



The screenshot shows a terminal window with the following content:

```

[hadoop@ip-172-31-5-80 ~]$ cat createHistorical.hql
CREATE EXTERNAL TABLE IF NOT EXISTS card_transactions_ext (
  card_id STRING,
  member_id STRING,
  amount DOUBLE,
  postcode STRING,
  pos_id STRING,
  transaction_dt STRING,
  status STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/hadoop/historical/card_transactions'
tblproperties("skip.header.line.count"="1");

LOAD DATA INPATH 'hdfs://user/hadoop/hive/card_transactions.csv' OVERWRITE INTO TABLE card_transactions_ext;

CREATE EXTERNAL TABLE IF NOT EXISTS card_transactions (row_key STRUCT<card_id:string, pos_id:string, transaction_dt:string>, card_id STRING, pos_id STRING, transaction_dt STRING, member_id STRING, amount DOUBLE, postcode STRING, status STRING)
ROW FORMAT DELIMITED
COLLECTION ITEMS TERMINATED BY '-'
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
WITH SERDEPROPERTIES ("hbase.columns.mapping"="TD:card_id, TD:pos_id, TD:transaction_dt,TD:member_id,TD:amount, TD:postcode, TD:status")
TBLPROPERTIES ("hbase.table.name" = "card_transactions");

Insert into card_transactions select NMED_STRUCT('card_id',card_id,'pos_id',pos_id,'transaction_dt', transaction_dt) as row_key,card_id,pos_id,transaction_dt, member_id, amount, postcode, status from card_transactions_ext;

CREATE VIEW IF NOT EXISTS last_ten_transactions
AS select card_id, member_id, amount, transaction_dt, postcode, rank() over (PARTITION BY card_id ORDER BY unix_timestamp(transaction_dt , 'dd-mm-yyyy hh:mm:ss') desc, amount desc) as ranking from card_transactions where status='GENUINE'
;

[hadoop@ip-172-31-5-80 ~]$ ls -lrt
total 8712
drwxr-xr-x 3 hadoop hadoop      149 Apr 27  2021 mysql-connector-java-8.0.25
-rw-rw-r-- 1 hadoop hadoop 4079310 Aug  7  2021 mysql-connector-java-8.0.25.tar.gz
-rw-rw-r-- 1 hadoop hadoop 4829520 Jan 17 07:48 card_transactions.csv
-rw-rw-r-- 1 hadoop hadoop    392 Jan 19 13:34 loadNosql.sh
-rw-rw-r-- 1 hadoop hadoop   1520 Jan 19 13:40 createHistorical.hql
[hadoop@ip-172-31-5-80 ~]$

```

```
=> ["card_transactions", "look_up_table"]
hbase(main):002:0> count 'card_transactions'
Current count: 1000, row: 341719092861087~348479157173653~09-12-2017 01:50:04~5777101.0
Current count: 2000, row: 343252340741735~878481175923264~17-03-2016 01:27:55~499306.0
Current count: 3000, row: 344453305077666~477889498690453~03-05-2017 13:17:03~4112459.0
Current count: 4000, row: 345864096406128~263597298380498~10-01-2018 13:19:13~5620846.0
Current count: 5000, row: 347177017703006~607111050994623~24-11-2017 19:29:35~3881266.0
Current count: 6000, row: 348390084708787~252122735905449~31-08-2017 15:58:36~4365570.0
Current count: 7000, row: 349261557797449~220369360742912~10-11-2017 13:09:04~4130551.0
Current count: 8000, row: 371085417506954~790236765996611~22-08-2016 15:36:09~3667024.0
Current count: 9000, row: 372838963275954~191667371054185~06-04-2017 08:12:05~5693002.0
Current count: 10000, row: 374437449333250~104804337714838~12-11-2016 17:23:01~9084333.0
Current count: 11000, row: 375956299108731~733657582169915~10-08-2017 23:08:32~9766446.0
Current count: 12000, row: 377037030364636~277121920702488~08-12-2017 13:32:44~3513692.0
Current count: 13000, row: 378118026791597~586336168666572~25-08-2017 19:31:43~1280928.0
Current count: 14000, row: 4009218272111551~298935629685938~01-08-2016 22:46:05~8656880.0
Current count: 15000, row: 4067184730430984~275178801561140~21-02-2017 14:21:14~6830136.0
Current count: 16000, row: 4132381122041426~598314604248219~10-12-2016 21:17:21~9303987.0
Current count: 17000, row: 4264553579186587~858567976578612~18-04-2017 23:55:12~6058344.0
Current count: 18000, row: 4373464339970856~657196939516344~11-01-2018 00:00:00~6559716.0
Current count: 19000, row: 4439501833420177~634293040387939~17-02-2017 05:01:05~5371581.0
Current count: 20000, row: 4500499651579063~277121920702488~08-12-2017 13:32:44~3513692.0
Current count: 21000, row: 4566853351752365~335349979924073~03-04-2017 20:30:40~3669382.0
Current count: 22000, row: 4600996294769125~259895316688347~25-12-2017 00:33:18~7070746.0
Current count: 23000, row: 4689314809377828~989600253556389~24-10-2016 02:52:59~185332.0
Current count: 24000, row: 4766789897935106~072938202866006~09-10-2017 20:58:38~4206082.0
Current count: 25000, row: 4863127030291206~483187165654021~12-05-2016 23:19:14~3878839.0
Current count: 26000, row: 4907253800863053~776565941881281~14-12-2017 09:59:35~7434309.0
Current count: 27000, row: 4995478705000641~155223326542744~20-05-2017 02:02:58~1780486.0
Current count: 28000, row: 5134479292018417~128069614660231~08-11-2017 08:29:00~4083279.0
Current count: 29000, row: 5162808285745682~924897469747000~02-07-2016 15:57:01~1888498.0
Current count: 30000, row: 5186615811954262~026744451387425~23-01-2018 06:47:40~2972350.0
Current count: 31000, row: 5221229593682054~127498352656607~16-08-2017 16:04:44~8870294.0
Current count: 32000, row: 5258095619226135~524892536444302~11-11-2017 00:00:00~5577468.0
Current count: 33000, row: 5294592751808411~698043686524445~28-12-2017 19:50:46~792320.0
Current count: 34000, row: 5342400571435088~569800919443173~26-08-2016 07:22:56~2049055.0
Current count: 35000, row: 5380978184175608~667233655872426~01-09-2016 10:51:09~9057468.0
Current count: 36000, row: 5414439899219272~345244546468970~07-06-2017 00:10:15~1413268.0
Current count: 37000, row: 5481808794715436~234167732114687~20-10-2017 16:48:20~3012181.0
Current count: 38000, row: 5534323829711423~637125777249291~30-06-2017 11:43:05~535263.0
Current count: 39000, row: 5584977018799504~770952296643375~03-05-2017 09:46:23~9635501.0
Current count: 40000, row: 6011139413319542~936287997923127~30-03-2016 07:13:43~1908818.0
Current count: 41000, row: 6011525010455848~959531034098755~02-08-2017 23:46:55~2762867.0
Current count: 42000, row: 6011782857327719~167569302823461~09-10-2016 02:59:08~8187310.0
Current count: 43000, row: 6221796595498984~174952353920145~18-02-2016 01:04:16~7689490.0
Current count: 44000, row: 6224271253849917~048019406374284~12-12-2017 19:22:13~4960131.0
Current count: 45000, row: 6225606551069826~648178895653883~13-07-2017 13:58:12~4112360.0
Current count: 46000, row: 6228733641419063~422091580968713~11-05-2016 10:27:11~2641410.0
Current count: 47000, row: 6447877814927926~992747968210744~11-01-2018 00:00:00~9774152.0
Current count: 48000, row: 6461356425954109~839304530643246~03-11-2016 20:57:23~1877568.0
Current count: 49000, row: 6480152634975473~963893207999520~17-01-2018 17:22:08~5863997.0
Current count: 50000, row: 6505080237250161~874636482279942~18-09-2017 17:36:33~2156682.0
Current count: 51000, row: 6544876671165176~424648847023528~04-08-2017 16:09:39~5155240.0
Current count: 52000, row: 6574255180086418~504711588075355~20-12-2017 08:11:48~5899422.0
Current count: 53000, row: 6595814135833988~311764663134170~18-11-2017 12:30:42~2717739.0
53292 row(s) in 2.2530 seconds

=> 53292
hbase(main):003:0> exit()
(hadoop@ip-172-31-5-175 ~)$ hive
```

```
[hadoop@ip-172-31-5-175 ~]$ hive
Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive> show tables;
OK
card_transactions
card_transactions_stg
last_ten_transactions
look_up_table
Time taken: 0.526 seconds, Fetched: 4 row(s)
hive> select count(*) from card_transactions;
Query ID = hadoop_20240117154216_87fbf425-ee48-4725-a348-1d670fabadbe
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705496453063_0012)
```

|           | VERTICES | MODE      | STATUS    | TOTAL | COMPLETED | RUNNING | PENDING | FAILED | KILLED |
|-----------|----------|-----------|-----------|-------|-----------|---------|---------|--------|--------|
| Map 1     | .....    | container | SUCCEEDED | 1     | 1         | 0       | 0       | 0      | 0      |
| Reducer 2 | .....    | container | SUCCEEDED | 1     | 1         | 0       | 0       | 0      | 0      |

VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 6.69 s

```
OK
53292
Time taken: 12.49 seconds, Fetched: 1 row(s)
hive> []
```

```
[hadoop@ip-172-31-5-80 ~]$ sh loadNosql.sh
rm: `hdfs:/user/hadoop/hive': No such file or directory
0 row(s) in 2.6260 seconds
```

```
Hbase::Table - card_transactions
0 row(s) in 2.6100 seconds
```

```
Hbase::Table - look_up_table
```

```
Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
OK
Time taken: 1.423 seconds
Loading data to table default.card_transactions_stg
OK
Time taken: 2.166 seconds
OK
Time taken: 1.293 seconds
Query ID = hadoop_20240119134254_4b355a4a-da7f-497b-a97f-e3f0526f6c2f
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705670818466_0002)
```

|       | VERTICES | MODE      | STATUS    | TOTAL | COMPLETED | RUNNING | PENDING | FAILED | KILLED |
|-------|----------|-----------|-----------|-------|-----------|---------|---------|--------|--------|
| Map 1 | .....    | container | SUCCEEDED | 1     | 1         | 0       | 0       | 0      | 0      |

VERTICES: 01/01 [=====>>>] 100% ELAPSED TIME: 6.79 s

```
OK
Time taken: 31.788 seconds
OK
Time taken: 0.343 seconds
[hadoop@ip-172-31-5-80 ~]$ []
```

## Task 2: Ingest the relevant data from AWS RDS to Hadoop using sqoop command.

```
[hadoop@ip-172-31-5-80 ~]$ cat dataIngestion.sh
#!/bin/bash

hdfs dfs -rm -r hdfs://user/hadoop/sqoop/import/

sqoop import --connect "jdbc:mysql://upgradawsrds1.cyaie1c9bmnf.us-east-1.rds.amazonaws.com:3306/cred_financials_data?connectionCollation=latin1_swedish_ci" \
--table card_member \
--target-dir /user/hadoop/sqoop/import/cred_financials_data/card_member \
--username upgraduser \
--m 1 \
--password upgraduser

sqoop import --connect "jdbc:mysql://upgradawsrds1.cyaie1c9bmnf.us-east-1.rds.amazonaws.com:3306/cred_financials_data?connectionCollation=latin1_swedish_ci" \
--table member_score \
--target-dir /user/hadoop/sqoop/import/cred_financials_data/member_score \
--username upgraduser \
--m 1 \
--password upgraduser

[hadoop@ip-172-31-5-80 ~]$
```

```
[hadoop@ip-172-31-5-80 ~]$ sh dataIngestion.sh
rm: hdfs://user/hadoop/sqoop/import/: No such file or directory
Warning: /usr/lib/sqoop/./accumulo does not exist! Accumulo imports will fail.
Please set HACCUMULO_HOME to the root of your Accumulo installation.
24/01/19 14:02:36 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/lib/hadoop/lib/slf4j-log4j12-1.7.10.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/share/awk/redisdb-jdbc-0.1.2.37.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/lib/hive/lib/log4j-slf4j-impl-2.6.2.jar/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
24/01/19 14:02:36 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
24/01/19 14:02:36 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
24/01/19 14:02:36 INFO tool.CodeGenTool: Beginning code generation
Loading class 'com.mysql.jdbc.Driver'. This is deprecated. The new driver class is 'com.mysql.cj.jdbc.Driver'. The driver is automatically registered via the SPI and manual loading of the driver class is generally unnecessary.
24/01/19 14:02:37 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'card_member' AS t LIMIT 1
24/01/19 14:02:37 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM 'card_member' AS t LIMIT 1
24/01/19 14:02:37 INFO org.apache.hadoop.mapred.lib.InputFormat: HADOOP_MAPRED_HOME is /usr/lib/hadoop-mapreduce
Note: /tmp/sqoop-hadoop/compile/b0d8b9c117d46241eaa9d2d236c7f9/card_member.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
24/01/19 14:02:39 INFO org.apache.hadoop.mapred.lib.InputFormat: Writing jar file: /tmp/sqoop-hadoop/compile/b0d8b9c117d46241eaa9d2d236c7f9/card_member.jar
24/01/19 14:02:39 WARN manager.MySQLManager: It looks like you are importing from mysql.
24/01/19 14:02:39 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
24/01/19 14:02:39 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
24/01/19 14:02:39 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
24/01/19 14:02:39 INFO mapreduce.ImportJobBase: Beginning import of card_member
24/01/19 14:02:40 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
24/01/19 14:02:40 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
24/01/19 14:02:40 INFO client.RMProxy: Connecting to ResourceManager at ip-172-31-5-80.ec2.internal:172.31.5.80:18032
24/01/19 14:02:48 INFO dh.DistributedFormat: Using read committed transaction isolation
24/01/19 14:02:48 INFO mapreduce.JobSubmitter: number of splits=1
24/01/19 14:02:48 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1705670818466_0005
24/01/19 14:02:48 INFO impl.YarnClientImpl: Submitted application application_1705670818466_0005
24/01/19 14:02:48 INFO mapreduce.Job: The url to track the job: http://ip-172-31-5-80.ec2.internal:20888/proxy/application_1705670818466_0005/
24/01/19 14:02:48 INFO mapreduce.Job: Running job: job_1705670818466_0005

24/01/18 08:25:31 INFO mapred.Task: Final Counters for attempt_local1899944623_0001_m_000000_0: Counters: 15
  File System Counters
    FILE: Number of bytes read=22390559
    FILE: Number of bytes written=23042119
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
  Map-Reduce Framework
    Map input records=999
    Map output records=999
    Input split bytes=87
    Spilled Records=0
    Failed Shuffles=0
    Merged Map outputs=0
    GC time elapsed (ms)=0
    Total committed heap usage (bytes)=459800576
  File Input Format Counters
    Bytes Read=0
  File Output Format Counters
    Bytes Written=85757
24/01/18 08:25:31 INFO mapred.LocalJobRunner: Finishing task: attempt_local1899944623_0001_m_000000_0
24/01/18 08:25:31 INFO mapred.LocalJobRunner: map task executor complete.
24/01/18 08:25:31 INFO mapreduce.Job: Job job_local1899944623_0001 running in uber mode : false
24/01/18 08:25:31 INFO mapreduce.Job: map 100% reduce 0%
24/01/18 08:25:31 INFO mapreduce.Job: Job job_local1899944623_0001 completed successfully
24/01/18 08:25:31 INFO mapreduce.Job: Counters: 15
  File System Counters
    FILE: Number of bytes read=22390559
    FILE: Number of bytes written=23042119
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
  Map-Reduce Framework
    Map input records=999
    Map output records=999
    Input split bytes=87
    Spilled Records=0
    Failed Shuffles=0
    Merged Map outputs=0
    GC time elapsed (ms)=0
    Total committed heap usage (bytes)=459800576
  File Input Format Counters
    Bytes Read=0
  File Output Format Counters
    Bytes Written=85757
24/01/18 08:25:31 INFO mapreduce.ImportJobBase: Transferred 0 bytes in 3.7118 seconds (0 bytes/sec)
24/01/18 08:25:31 INFO mapreduce.ImportJobBase: Retrieved 999 records.
```

```
Note: /tmp/sqoop-root/compile/cfbdf063fae7d60267489cf1b17bf0b4/member_score.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
24/01/18 08:32:08 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-root/compile/cfbdf063fae7d60267489cf1b17bf0b4/member_score.jar
24/01/18 08:32:08 WARN manager.MySQLManager: It looks like you are importing from mysql.
24/01/18 08:32:08 WARN manager.MySQLManager: This transfer can be faster! Use the --direct
24/01/18 08:32:08 WARN manager.MySQLManager: option to exercise a MySQL-specific fast path.
24/01/18 08:32:08 INFO manager.MySQLManager: Setting zero DATETIME behavior to convertToNull (mysql)
24/01/18 08:32:08 INFO mapreduce.ImportJobBase: Beginning import of member_score
24/01/18 08:32:08 INFO Configuration.deprecation: mapred.jar is deprecated. Instead, use mapreduce.job.jar
24/01/18 08:32:09 INFO Configuration.deprecation: mapred.map.tasks is deprecated. Instead, use mapreduce.job.maps
24/01/18 08:32:09 INFO client.RMProxy: Connecting to ResourceManager at ip-172-31-4-180.ec2.internal/172.31.4.180:8032
24/01/18 08:32:19 INFO db.DBInputFormat: Using read committed transaction isolation
24/01/18 08:32:19 INFO mapreduce.JobSubmitter: number of splits=1
24/01/18 08:32:19 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1705566423041_0001
24/01/18 08:32:20 INFO impl.YarnClientImpl: Submitted application application_1705566423041_0001
24/01/18 08:32:20 INFO mapreduce.Job: The url to track the job: http://ip-172-31-4-180.ec2.internal:20888/proxy/application_1705566423041_0001/
24/01/18 08:32:20 INFO mapreduce.Job: Running job: job_1705566423041_0001
24/01/18 08:32:27 INFO mapreduce.Job: Job job_1705566423041_0001 running in uber mode : false
24/01/18 08:32:27 INFO mapreduce.Job: map 0% reduce 0%
24/01/18 08:32:34 INFO mapreduce.Job: map 100% reduce 0%
24/01/18 08:32:35 INFO mapreduce.Job: Job job_1705566423041_0001 completed successfully
24/01/18 08:32:35 INFO mapreduce.Job: Counters: 30
File System Counters
  FILE: Number of bytes read=0
  FILE: Number of bytes written=189753
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=87
  HDFS: Number of bytes written=19980
  HDFS: Number of read operations=4
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
Job Counters
  Launched map tasks=1
  Other local map tasks=1
  Total time spent by all maps in occupied slots (ms)=376608
  Total time spent by all reduces in occupied slots (ms)=0
  Total time spent by all map tasks (ms)=3923
  Total vcore-milliseconds taken by all map tasks=3923
  Total megabyte-milliseconds taken by all map tasks=12051456
Map-Reduce Framework
  Map input records=999
  Map output records=999
  Input split bytes=87
  Spilled Records=0
  Failed Shuffles=0
  Merged Map outputs=0
  GC time elapsed (ms)=61
  CPU time spent (ms)=1700
  Physical memory (bytes) snapshot=306360320
  Virtual memory (bytes) snapshot=4613513216
  Total committed heap usage (bytes)=311951360
File Input Format Counters
  Bytes Read=0
File Output Format Counters
  Bytes Written=19980
24/01/18 08:32:35 INFO mapreduce.ImportJobBase: Transferred 19.5117 KB in 26.531 seconds (753.0805 bytes/sec)
24/01/18 08:32:35 INFO mapreduce.ImportJobBase: Retrieved 999 records.
[hadoop@ip-172-31-4-180 ~]$ hadoop fs -ls /user/hadoop/sqoop/import/cred_financials_data/card_member/
Found 2 items
-rw-r--r-- 1 root hadoop 0 2024-01-18 08:42 /user/hadoop/sqoop/import/cred_financials_data/card_member/_SUCCESS
-rw-r--r-- 1 root hadoop 85081 2024-01-18 08:42 /user/hadoop/sqoop/import/cred_financials_data/card_member/part-m-00000
[hadoop@ip-172-31-4-180 ~]$
[hadoop@ip-172-31-4-180 ~]$ hadoop dfs -ls sqoop/import/cred_financials_data/member_score
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
Found 2 items
-rw-r--r-- 1 root hadoop 0 2024-01-18 08:32 sqoop/import/cred_financials_data/member_score/_SUCCESS
-rw-r--r-- 1 root hadoop 19980 2024-01-18 08:32 sqoop/import/cred_financials_data/member_score/part-m-00000
[hadoop@ip-172-31-4-180 ~]$
```

No of rows imported in each file: 999

```
[hadoop@ip-172-31-4-180 ~]$ hadoop fs -cat /user/hadoop/sqoop/import/cred_financials_data/card_member/part-m-00000 | wc -l
999
[hadoop@ip-172-31-4-180 ~]$ hadoop fs -cat sqoop/import/cred_financials_data/member_score/part-m-00000 | wc -l
999
[hadoop@ip-172-31-4-180 ~]$
```

**Task 3:** Create a look-up table with columns specified earlier in the problem statement.

```
[hadoop@ip-172-31-5-80 ~]$ cat createLookup.sh
#!/bin/bash

echo "create 'look_up_table', 'card_details', 'Member_details', 'Location', 'Rule_params'" | hbase shell -n
hive -f /home/hadoop/lookuptable.hql
[hadoop@ip-172-31-5-80 ~]$
```

```
[hadoop@ip-172-31-5-80 ~]$ cat lookuptable.hql
CREATE EXTERNAL TABLE IF NOT EXISTS look_up_table (
  card_id STRING,
  card_purchase_dt STRING,
  transaction_dt STRING,
  member_id STRING,
  member_joining_dt STRING,
  country STRING,
  city STRING,
  UCL DOUBLE,
  postcode STRING,
  score INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED BY 'org.apache.hadoop.hive.hbase.HBaseStorageHandler'
WITH SERDEPROPERTIES ("hbase.columns.mapping" = "card_details:card_purchase_dt,card_details:transaction_dt,Member_details:member_id,Member_details:member_joining_dt,Location:country,Location:city,Rule_params:UCL,Rule_params:Rule_params:score")
TBLPROPERTIES ("hbase.table.name" = "look_up_table");
[hadoop@ip-172-31-5-80 ~]$
```

```
[hadoop@ip-172-31-5-80 ~]$ sh createLookup.sh

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
OK
Time taken: 2.315 seconds
[hadoop@ip-172-31-5-80 ~]$
```

```
[hadoop@ip-172-31-5-80 ~]$ sudo hbase shell
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
Version 1.4.13, rUnknown, Fri Apr 17 15:18:24 UTC 2020

hbase(main):001:0> list
TABLE
card_transactions
look_up_table
2 row(s) in 0.1570 seconds

=> ["card_transactions", "look_up_table"]
hbase(main):002:0> scan "look_up_table"
ROW                                     COLUMN+CELL
0 row(s) in 0.0800 seconds

hbase(main):003:0>
```

```
[hadoop@ip-172-31-5-80 ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive> show tables;
OK
card_transactions
card_transactions_stg
last_ten_transactions
look_up_table
Time taken: 0.504 seconds, Fetched: 4 row(s)
hive> select count(*) from look_up_table;
OK
0
Time taken: 1.746 seconds, Fetched: 1 row(s)
hive>
```

**Task 4:** After creating the table, you need to load the relevant data in the lookup table.

```
[hadoop@ip-172-31-5-80 ~]$ vi preAnalysis.shql
[hadoop@ip-172-31-5-80 ~]$ cat preAnalysis.hql
CREATE EXTERNAL TABLE IF NOT EXISTS card_member (
  card_id STRING,
  member_id STRING,
  member_joining_dt STRING,
  card_purchase_dt STRING,
  country STRING,
  city STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/hadoop/RDS/card_member';

CREATE EXTERNAL TABLE IF NOT EXISTS member_score (
  member_id STRING,
  score INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/hadoop/RDS/member_score';

LOAD DATA INPATH 'hdfs://user/hadoop/sqoop/import/cred_financials_data/card_member' OVERWRITE INTO TABLE card_member;
LOAD DATA INPATH 'hdfs://user/hadoop/sqoop/import/cred_financials_data/member_score' OVERWRITE INTO TABLE member_score;

Insert into look_up_table select ltt.card_id, cm.card_purchase_dt, ltt.transaction_dt, ltt.member_id, member_joining_dt, country, city, UCL, ltt.postcode, score
from last_ten_transactions ltt
inner join
member_score ms
on ltt.member_id=ms.member_id and ltt.ranking=1
inner join
card_member cm
on cm.member_id=ltt.member_id and ltt.ranking=1
inner join
(select card_id, avg(amount)+ (3* stddev(amount)) as UCL
from last_ten_transactions
where ranking<=10
group by card_id
) as ucl
on ltt.card_id=ucl.card_id ;

exit;
```

```
[hadoop@ip-172-31-5-80 ~]$ hive -f preAnalysis.hql

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
OK
Time taken: 0.912 seconds
OK
Time taken: 0.038 seconds
Loading data to table default.card_member
OK
Time taken: 0.702 seconds
Loading data to table default.member_score
OK
Time taken: 0.367 seconds
No Stats for default@card_transactions, Columns: member_id, amount, postcode, transaction_dt, card_id, status
No Stats for default@member_score, Columns: member_id, score
No Stats for default@card_member, Columns: member_id, country, member_joining_dt, city, card_purchase_dt
No Stats for default@card_transactions, Columns: amount, transaction_dt, card_id, status
Query ID = hadoop_20240119142540_7f492c8c-a6fb-4889-a504-acc882380a9a
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705670818466_0010)

-----
      VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Map 3 ..... container  SUCCEEDED    1         1         0         0         0         0
Map 4 ..... container  SUCCEEDED    1         1         0         0         0         0
Map 5 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    2         2         0         0         0         0
Reducer 6 ..... container  SUCCEEDED    2         2         0         0         0         0
Reducer 7 ..... container  SUCCEEDED    2         2         0         0         0         0
-----
VERTICES: 07/07 [=====>>>] 100% ELAPSED TIME: 12.10 s
-----
OK
Time taken: 20.524 seconds
hadoop@ip-172-31-5-80 ~$
```

```
[hadoop@ip-172-31-5-80 ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: true
hive> select count(*) from card_member;
Query ID = hadoop_20240119142951_e6f594f2-87bc-4caa-8b95-2da2a6a9a6fd
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705670818466_0011)

-----
      VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 4.57 s
-----
OK
999
Time taken: 8.079 seconds, Fetched: 1 row(s)
hive> select count(*) from member_score;
Query ID = hadoop_20240119143003_54422958-8540-4aad-b3ce-070cbf31e7c0
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1705670818466_0011)

-----
      VERTICES      MODE      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 ..... container  SUCCEEDED    1         1         0         0         0         0
Reducer 2 ..... container  SUCCEEDED    1         1         0         0         0         0
-----
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 0.39 s
-----
OK
999
Time taken: 0.892 seconds, Fetched: 1 row(s)
hive>
```

```

6595814135833988 column=Rule_params:UCL, timestamp=1705674360996, value=1.3926273240525039E7
6595814135833988 column=Rule_params:postcode, timestamp=1705674360996, value=22508
6595814135833988 column=Rule_params:score, timestamp=1705674360996, value=210
6595814135833988 column=card_details:card_purchase_dt, timestamp=1705674360996, value=06/14
6595814135833988 column=card_details:transaction_dt, timestamp=1705674360996, value=30-01-2018 02:03:54
6595928469079750 column=Location:city, timestamp=1705674360996, value=Raleigh
6595928469079750 column=Location:country, timestamp=1705674360996, value=United States
6595928469079750 column=Member_details:member_id, timestamp=1705674360996, value=422173403853722
6595928469079750 column=Member_details:member_joining_dt, timestamp=1705674360996, value=2012-02-10 02:19:20.0
6595928469079750 column=Rule_params:UCL, timestamp=1705674360996, value=1.142787041440079E7
6595928469079750 column=Rule_params:postcode, timestamp=1705674360996, value=98349
6595928469079750 column=Rule_params:score, timestamp=1705674360996, value=412
6595928469079750 column=card_details:card_purchase_dt, timestamp=1705674360996, value=02/14
6595928469079750 column=card_details:transaction_dt, timestamp=1705674360996, value=24-01-2018 12:38:22
6597703848279563 column=Location:city, timestamp=1705674360996, value=Pinewood
6597703848279563 column=Location:country, timestamp=1705674360996, value=United States
6597703848279563 column=Member_details:member_id, timestamp=1705674360996, value=657991752750272
6597703848279563 column=Member_details:member_joining_dt, timestamp=1705674360996, value=2017-02-16 07:45:34.0
6597703848279563 column=Rule_params:UCL, timestamp=1705674360996, value=1.471863414949845E7
6597703848279563 column=Rule_params:postcode, timestamp=1705674360996, value=95699
6597703848279563 column=Rule_params:score, timestamp=1705674360996, value=218
6597703848279563 column=card_details:card_purchase_dt, timestamp=1705674360996, value=03/17
6597703848279563 column=card_details:transaction_dt, timestamp=1705674360996, value=27-01-2018 10:51:49
6598830758632447 column=Location:city, timestamp=1705674360996, value=Tampa
6598830758632447 column=Location:country, timestamp=1705674360996, value=United States
6598830758632447 column=Member_details:member_id, timestamp=1705674360996, value=660350842563890
6598830758632447 column=Member_details:member_joining_dt, timestamp=1705674360996, value=2013-07-26 02:22:17.0
6598830758632447 column=Rule_params:UCL, timestamp=1705674360996, value=1.2227849982601807E7
6598830758632447 column=Rule_params:postcode, timestamp=1705674360996, value=19421
6598830758632447 column=Rule_params:score, timestamp=1705674360996, value=293
6598830758632447 column=card_details:card_purchase_dt, timestamp=1705674360996, value=12/16
6598830758632447 column=card_details:transaction_dt, timestamp=1705674360996, value=30-01-2018 00:18:34
6599900931314251 column=Location:city, timestamp=1705674360996, value=Loves Park
6599900931314251 column=Location:country, timestamp=1705674360996, value=United States
6599900931314251 column=Member_details:member_id, timestamp=1705674360996, value=928036864799687
6599900931314251 column=Member_details:member_joining_dt, timestamp=1705674360996, value=2017-09-09 03:31:54.0
6599900931314251 column=Rule_params:UCL, timestamp=1705674360996, value=1.2121408572464656E7
6599900931314251 column=Rule_params:postcode, timestamp=1705674360996, value=97423
6599900931314251 column=Rule_params:score, timestamp=1705674360996, value=297
6599900931314251 column=card_details:card_purchase_dt, timestamp=1705674360996, value=10/17
6599900931314251 column=card_details:transaction_dt, timestamp=1705674360996, value=31-01-2018 11:25:16
999 row(s) in 10.8880 seconds

```

```

6599900931314251
999 row(s) in 10.8880 seconds

hbase(main):004:0> count 'look_up_table'
999 row(s) in 0.1110 seconds

=> 999
hbase(main):005:0> █

```